

AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2019

Physics Paper I

Time: 50 minutes Marks: 35

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 35 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.

Correct Way	Incorrect Ways
1 ☐ A ☐ B ☒ C ☐ D	1 ☐ A ☐ B ☒ C ☐ D
	2 ☐ A ☐ B ☒ C ☐ D
	3 ☐ A ☐ B ☒ C ☐ D
	4 ☐ A ☐ B ☒ C ☐ D

Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. The option that shows three base quantities' units in System International (SI) is
 - A. gram, joule and second.
 - B. gram, centimetre and dyne.
 - C. kilogram, metre and second.
 - D. kilogram, newton and second.
2. The second equation of motion of a uniformly accelerated ball along a straight line is stated as
$$s = v_i t + \frac{1}{2} a t^2 .$$

The dimension of this equation is

- A. [L]
 - B. [T]
 - C. [L][T⁻¹]
 - D. [L][T⁻²]
3. If the tail of the first vector coincides with the head of the second vector, the resultant of two vectors will be found by
 - A. division of vectors.
 - B. addition of vectors.
 - C. subtraction of vectors.
 - D. multiplication of vectors.
4. If a vector is multiplied by a negative number, then its direction will
 - A. be reversed.
 - B. be changed by 45°.
 - C. be changed by 90°.
 - D. remain the same.
5. If both the components of a vector are negative, then the vector lies in the
 - A. first quadrant.
 - B. second quadrant.
 - C. third quadrant.
 - D. fourth quadrant.
6. The vector product of moment arm $\vec{r}$ and force $\vec{F}$ is called torque.

The direction of torque is

 - A. along the force $\vec{F}$.
 - B. along the moment arm $\vec{r}$.
 - C. parallel to the plane containing $\vec{r}$ and $\vec{F}$.
 - D. perpendicular to the plane containing $\vec{r}$ and $\vec{F}$.

7. A paratrooper falling with uniform velocity is an example of

- A. static equilibrium.
- B. stable equilibrium.
- C. unstable equilibrium.
- D. dynamic equilibrium.

8. An object falls freely from rest. The total distance covered by it in 2 s will be

(Note: The acceleration due to gravity 'g' is 9.8 m/s^2 .)

- A. 4.9 m.
- B. 19.6 m.
- C. 29.4 m.
- D. 39.2 m.

9. A football kicked by a football player in the air is an example of

- A. linear motion.
- B. circular motion.
- C. projectile motion.
- D. rotational motion.

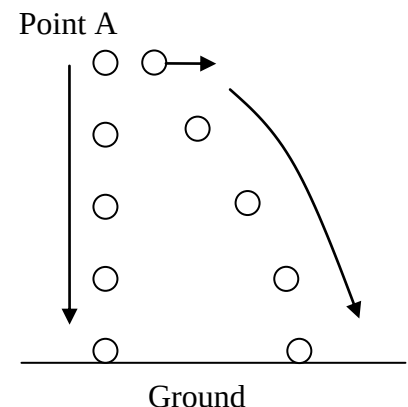


10. When a projectile reaches the highest point of the trajectory, the vertical component of its velocity becomes

- A. zero.
- B. half.
- C. double.
- D. maximum.

11. Two balls are released simultaneously from a point 'A' with zero vertical velocity as shown in the given diagram. These balls followed different paths but reached the ground at the same time due to the

- A. same size of balls.
- B. horizontal velocity.
- C. same vertical height.
- D. horizontal acceleration.



12. The work done by the gravitational force on a body in displacing it from a given position near the surface of Earth to infinity is called

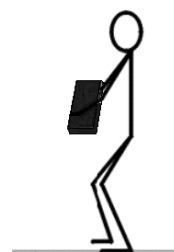
I. elastic potential energy.
II. potential energy of the body.
III. absolute potential energy of the body.

A. I only
B. III only
C. I and II
D. II and III

13. The given diagram represents a boy lifting a block in the upward direction.

The work done by the gravitational force on the block is

A. zero.
B. positive.
C. variable.
D. negative.



14. When an object moves with uniform speed in a circular orbit, its centripetal acceleration will be directed

A. along the circumference of circle.
B. along the direction of motion of the object.
C. towards the centre of the circle along the radius.
D. away from the centre of the circle along the radius.

15. In all of the following sports, law of conservation of angular momentum is used EXCEPT in

A. ice-skating.
B. gymnastics.
C. horse riding.
D. board diving.

16. In a cement factory, a viscous liquid flows through a pipe with an area of cross-section 6 m^2 at a velocity of 8 m/s .

When this fluid moves forward, the area of cross-section of the pipe decreases to 4 m^2 . The velocity of the liquid will now be

A. 3 m/s .
B. 12 m/s .
C. 18 m/s .
D. 24 m/s .

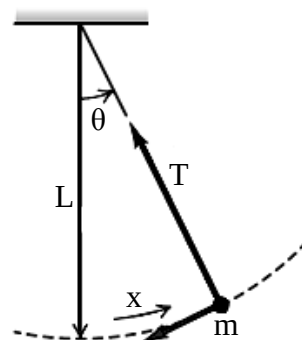
17. Bernoulli's equation is the fundamental equation in fluid dynamics that relates pressure of fluid to its speed and height.

According to the Bernoulli's equation, where the speed of the fluid is high, the pressure will be

- A. low.
 - B. zero.
 - C. high.
 - D. constant.
18. All of the following are the factors on which terminal velocity of a falling object depends EXCEPT
- A. mass of the object.
 - B. velocity of the object.
 - C. cross-sectional area of the object.
 - D. gravitational acceleration of the object.
19. A body is moving along a circular path. The projection of the body along the diameter of the circle will execute
- A. angular motion.
 - B. rotatory motion.
 - C. circulatory motion.
 - D. simple harmonic motion.
20. If the length of a simple pendulum becomes four times, then its time period will become
- A. two times.
 - B. four times.
 - C. six times.
 - D. eight times.
21. The given figure shows a simple pendulum executing simple harmonic motion (SHM).

The restoring force in the given simple pendulum is equal to

- A. $-mg\theta$
- B. $-mg$
- C. $mg\theta$
- D. mg



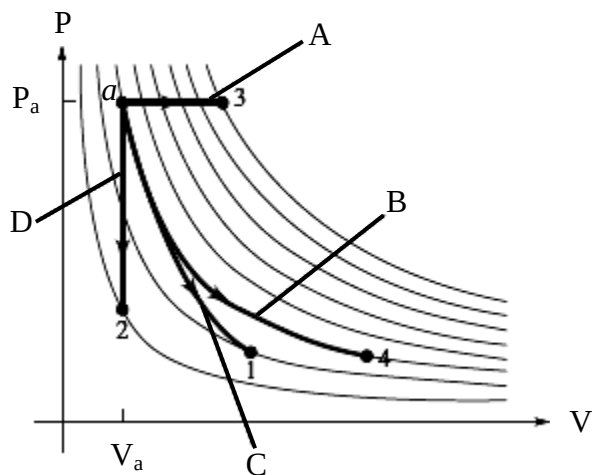
22. If 20 sound waves pass from a point of a medium in 2 s with the velocity of 20 m/s, then the wavelength is
- A. 1 m.
 - B. 2 m.
 - C. 10 m.
 - D. 20 m.

23. The speed of sound wave in a medium depends upon
- I. mass of the medium
 - II. density of the medium
 - III. elasticity of the medium
- A. I only.
 - B. III only.
 - C. I and II.
 - D. II and III.
24. Which of the following properties of sound is affected by the change in temperature of the atmosphere?
- I. Velocity
 - II. Frequency
 - III. Wavelength
- A. I only.
 - B. III only.
 - C. I and II.
 - D. II and III.
25. In a stationary wave, the velocity of particles of a medium at the nodes
- A. becomes zero.
 - B. becomes maximum.
 - C. varies unpredictably.
 - D. remains constant but non-zero.
26. When a fire brigade approaches to a stationary listener, the frequency of sound of its siren received by the listener
- A. varies unpredictably.
 - B. remains the same as the original sound.
 - C. becomes less than the frequency of original sound.
 - D. becomes greater than the frequency of original sound.
27. When two or more waves overlap at the same time the resultant wave will be the sum of individual waves. The principle is known as
- A. reflection.
 - B. ultrasonic.
 - C. supersonic.
 - D. superposition.
28. The phenomenon of wave that distinguishes between transverse and longitudinal wave is
- A. refraction.
 - B. diffraction.
 - C. interference.
 - D. polarisation.

29. Which of the following is NOT the application of polarisation of light?
- A. Photographers use filters to reduce glare in order to get clear pictures.
 - B. Drivers use sunglasses that diminish the brightness of headlights of a car.
 - C. Fishermen use eye-wear which decreases the reflection of light from water.
 - D. Astronomers use lenses in telescope to enlarge the image of celestial bodies.
30. In an isothermal process, the internal energy of the system
- A. increases.
 - B. decreases.
 - C. remains constant.
 - D. varies unpredictably.
31. The second law of thermodynamics deals with conversion of heat into work and direction of
- A. flow of heat.
 - B. applied force.
 - C. applied pressure.
 - D. flow of internal energy.
32. In Carnot cycle, at the end of the cyclic path, the final temperature of the working substance will be
- I. higher than the initial temperature
 - II. lower than the initial temperature
 - III. equal to the initial temperature
- A. I only.
 - B. III only.
 - C. I or III.
 - D. II or III.
33. The efficiency of Carnot engine depends upon temperature of the
- A. sink only.
 - B. source only.
 - C. surrounding.
 - D. source and sink.
34. When an ideal gas is cooled at constant volume until the pressure becomes half of the original pressure, then the average translational kinetic energy of the gas molecules
- A. remains unchanged.
 - B. becomes half the original.
 - C. becomes twice the original.
 - D. becomes thrice the original.

35. In the given figure, four different processes for a constant amount of an ideal gas, all starting at state a , are shown.

In which of the given processes, the system does ZERO work on its surrounding?



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